Penetration Test Report

MedTech Solutions

Security Assessment Findings Report

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1. Confidentiality Statement

This document contains proprietary and confidential information belonging to MedTech Solutions and CyberSecure Consulting Group (CCG). Unauthorized copying, disclosure, or distribution is strictly prohibited. Access is limited to authorized personnel only.

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2. Disclaimer

This penetration test represents the security posture of MedTech Solutions during the assessment period of **May 15, 2024 – May 22, 2024**. Any system changes made after the assessment may affect the validity of these findings.

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3. Contact Information

Name	Title	Contact Information
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Innocent Utu	Junior Penetration Tester (Report Author)	Student Lab Submission

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4. Assessment Overview

The purpose of this assessment was to evaluate the security posture of **MedTech Solutions** by identifying vulnerabilities, security weaknesses, and configuration issues affecting its external infrastructure, cloud environment, and web applications.

The engagement was conducted from **May 15, 2024 to May 22, 2024** using industry-recognized methodologies including:

- NIST SP 800-115 (Technical Guide to Information Security Testing)
- OWASP Testing Guide
- MITRE ATT&CK Framework

Testing Phases

Planning

- Defined engagement objectives
- Established rules of engagement

Discovery

- Performed reconnaissance
- Network scanning
- Enumeration
- OSINT gathering

Attack

- Exploited discovered vulnerabilities
- Tested authentication mechanisms
- Conducted API testing
- Performed privilege escalation testing

Reporting

- Documented vulnerabilities
- Assigned severity ratings
- Recommended remediation actions

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5. Assessment Components

Assessment Type	Description
External Penetration Test	Evaluated publicly accessible services including patient login portals and remote access services.
Internal Penetration Test	Not included in this engagement.
Web Application Testing	Tested the EMR application and patient portal for SQL injection, authentication flaws, and session management weaknesses.
Cloud Security Review	Assessed AWS S3 storage, API endpoints, database access controls, and cloud configurations.

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6. Finding Severity Ratings

Severity	CVSS Score	Definition
Critical	9.0–10.0	Immediate action required due to high risk of system compromise.
High	7.0–8.9	Serious vulnerability that could lead to unauthorized access or privilege escalation.
Medium	4.0–6.9	Vulnerability requiring additional factors for successful exploitation.
Low	0.1–3.9	Minor security concern.
Informational	N/A	Security observations with no immediate risk.

Findings Summary

Severity	Number of Findings
Critical	2
High	3
Medium	4
Low	0

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7. Scope

In-Scope Assets

- EMR Web Application
- Patient Portal
- AWS S3 Cloud Storage
- API Endpoints
- Database Servers

Scope Exclusions

- Denial of Service (DoS) attacks
- Phishing attacks
- Social engineering attacks

Client Allowances

- Authorized penetration testing during the assessment period.
- Permission to use automated scanning and exploitation tools.
- Permission to test cloud infrastructure and APIs.

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8. Executive Summary

The **MedTech Solutions** penetration test identified **2 Critical**, **3 High**, and **4 Medium** severity vulnerabilities.

The most significant issues were insecure API authentication and weak database access controls. These vulnerabilities could allow attackers to gain unauthorized access to sensitive patient records and proprietary medical information.

The assessment also showed positive security practices, including structured cloud infrastructure and adherence to recognized testing standards. However, weaknesses in authentication, authorization, and access control significantly increase the organization's risk of data breaches and regulatory violations.

Immediate remediation of Critical and High-risk findings is strongly recommended.

Attack Summary

Step	Action	Recommendation
1	Performed reconnaissance and scanned public-facing systems using Nmap.	Limit exposed services and continuously monitor external assets.
2	Tested API authentication and identified unsecured endpoints exposing patient data.	Enforce authentication and authorization on all APIs.
3	Exploited default database credentials to access sensitive medical records.	Remove default credentials, implement MFA, and enforce strong access controls.

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9. Security Strengths

The assessment identified several effective security controls:

- Testing followed NIST, OWASP, and MITRE ATT&CK standards.
- Cloud infrastructure was organized and clearly defined.
- Multiple security assessment techniques were utilized.
- Comprehensive testing covered networks, web applications, and cloud resources.
- Security assessment included manual and automated testing techniques.

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10. Security Weaknesses

The following weaknesses were identified:

- Missing authentication on API endpoints.
- Weak database access controls using default credentials.
- Authentication mechanisms vulnerable to brute-force attacks.
- Cloud access control misconfigurations.
- Session management weaknesses.
- Insufficient authorization controls protecting patient records.

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11. Vulnerabilities by Impact

Vulnerability	Severity	Impact	Affected Systems	Recommendation
Insecure API Authentication (CVSS 9.5)	Critical	Unauthorized users could retrieve patient records without authentication.	API Endpoints	Enforce authentication using OAuth 2.0 or JWT and validate authorization on every request.
Weak Database Access Controls (CVSS 9.2)	Critical	Default credentials allowed unauthorized access and	Database Servers	Remove default credentials, implement MFA, and enforce least privilege access.

		modification of medical records.		
Authentication Weaknesses	High	Increased risk of unauthorized account access.	Patient Portal	Strengthen password policy, enable MFA, and implement account lockout.
Cloud Configuration Issues	High	Potential exposure of cloud-hosted patient data.	AWS S3 Environment	Restrict IAM permissions, review bucket policies, and perform regular cloud audits.
Session Management Weaknesses	High	Increased risk of session hijacking.	EMR Web Application	Use secure cookies, shorter session timeouts, and session token protection.
Security Misconfigurations	Medium	Increased attack surface.	Multiple Systems	Apply secure configuration baselines.
Information Disclosure	Medium	Sensitive information exposed to attackers.	Web Application	Limit error messages and sensitive responses.
Weak Security Headers	Medium	Browser-based attacks become easier.	Patient Portal	Implement recommended HTTP security headers.
Minor Access Control Issues	Medium	Unauthorized users may access limited resources.	Cloud Services	Review permissions and apply least privilege principles.

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12. Additional Reports and Scans

The following supporting documents would normally accompany this report:

- Findings_Report.xlsx – Detailed vulnerability findings with CVSS scoring.
- Vulnerability_Scan_Report.pdf – Full vulnerability scan results.
- Attack_Path_Analysis.docx – Detailed attack methodology and exploitation paths.

Conclusion & Next Steps

The penetration test demonstrated that MedTech Solutions has several critical security vulnerabilities that could expose sensitive patient information and proprietary medical systems if left unaddressed.

The organization should prioritize remediation of Critical and High severity vulnerabilities by implementing secure authentication mechanisms, strengthening database access controls, enforcing Multi-Factor Authentication (MFA), and improving cloud security configurations.

Next Steps

- Immediate remediation of all Critical vulnerabilities.
- Secure API authentication and authorization.
- Remove default database credentials.
- Implement Multi-Factor Authentication (MFA).
- Review AWS cloud permissions and IAM policies.
- Perform regular vulnerability assessments and penetration tests.
- Enhance continuous monitoring and logging.
- Provide ongoing security awareness training for employees.